

In this case, we have a basic trading model which appears to have positive expected returns over time. Knowing that, we can decide how high those returns should be, simply by changing the position-sizing mechanism. All we need to do is to take on larger positions and, hey, presto!, higher returns. Too bad that reality isn't as cooperative.

Remember how we use a position-sizing methodology based on Average True Range, with a target daily variation of 15 basis points? This is the key number to play with. For that reason, we're going to play with some variations.

I'm going to make one riskier version, and two less risky variants. Our base variant so far is using 15 basis points as the target daily variation, and I will show you versions based on 30 points, 10, and 7.5.

As you see in Table 4.5, we could crank the risk up to 30 points and get a return of nearly 26% per year. This is of course a very dangerous illusion and unlikely to be achieved in real life. Would you really want to put serious money at work on a strategy which would see losses of over 65% at some point? Always assume that real-life results will be worse than a backtest, so you might actually see losses of 75%, 80%, or more.

Here's a piece of simple maths to illustrate why a drawdown of 70% is a very bad idea. As simple as this example is, it is surprisingly often overlooked in the retail trading community. Imagine starting out the year with \$100,000, and suffering a loss of 70%. You now have \$30,000. If the next year returns +70%, you're not back to where you started. No, you'd still have a near 50% loss. $\$30,000 \times 1.7 = \$51,000$.

TABLE 4.5 Controlling the risk level

	Core model 7.5bp	Core model 10bp	Core model 15bp	Core model 30bp
Compounded annual return	8.88	11.63	15.81	25.94
Worst drawdown	-22.23	-28.69	-39.43	-64.66
Annualised volatility	13.08	17.41	25.90	49.57
Percentage profitable months	58.30	58.80	57.10	57.50
Best month	17.30	23.50	37.00	69.30
Worst month	-9.00	-12.20	-18.37	-35.90
Sharpe (zero)	0.72	0.72	0.72	0.71